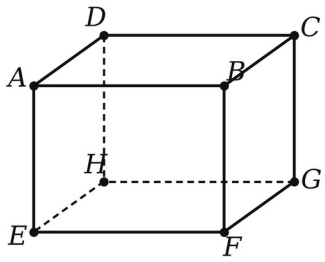
Honors Geometry – Ch. 3 Test PRACTICE Full Name: _____

Parallel and Perpendicular Lines • Sections 3.1 – 3.5 Period: _____ _____ / 40

Show all work in the space provided. Check your work. Write each final answer on the line. Diagrams are not drawn to scale.

Use the box for questions 1 – 3.



1. Name a line parallel to line DH. (1)

_____

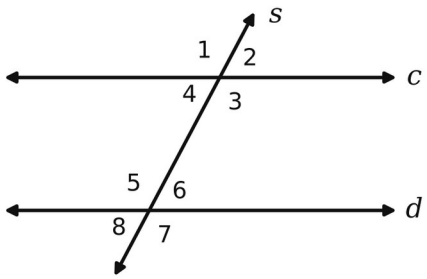
2. Name a line skew to line AD. (1)

_____

3. Name a plane parallel to plane ABF. (1)

_____

Use the diagram for questions 4 – 9. In 7 – 9, c ∥ d.



4. Classify the angle pair: ∠1 and ∠5. (1)

_____

5. Classify the angle pair: ∠6 and ∠4. (1)

_____

6. Classify the angle pair: ∠3 and ∠6. (1)

_____

7. m∠7 = 118°. Find m∠1. (2)

m∠1 = _____

8. m∠3 = 121°. Find m∠6. (2)

m∠6 = _____

9. m∠5 = 116°. Find m∠2. (2)

m∠2 = _____

10. Two parallel lines are cut by a transversal. A pair of corresponding angles measure (5x – 7)° and 68°. Find x. (2)

x = _____

11. Two parallel lines are cut by a transversal. A pair of consecutive interior angles measure (4x + 6)° and (3x + 13)°. Find x and BOTH angle measures. (2)

x = _____ _____° and _____°

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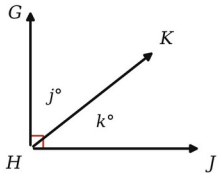
12. Use the diagram for 4 – 9, but suppose you do NOT know that c ∥ d. If ∠4 and ∠5 are supplementary, then c ∥ d. Name the theorem that proves it. (2)

_____

13. Two lines are cut by a transversal. A pair of alternate interior angles measure (7x − 4)° and (4x + 20)°. Find the value of x that makes the lines parallel. (2)

x = _____

14. In the diagram, GH ⊥ HJ. j = (2x + 16) and k = (4x − 10). Find x. (2)



x = _____

15. In a plane, line s ⊥ line u and line t ⊥ line u. What can you conclude about lines s and t? Name the theorem. (2)

_____

16. In a plane, line c ∥ line d and line e ⊥ line c. What can you conclude about lines e and d? Name the theorem. (2)

_____

17. Point P lies on the perpendicular bisector of AB. PA = 4x + 1 and PB = 6x − 9. Find PA. (2)

PA = _____

18. Find the slope of the line through (−2, 7) and (6, 1). (1)

m = _____

19. Parallel, perpendicular, coinciding, or neither? Explain. $y = \frac{1}{3}x + 4$ and $x - 3y = 9$ (2)

_____

20. Write an equation in slope-intercept form of the line through (3, −1) that is PARALLEL to y = 4x − 2. (2)

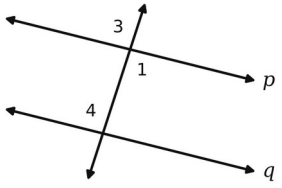
y = _____

21. Write an equation in slope-intercept form of the line through (2, 7) that is PERPENDICULAR to x − 4y = 8. (2)

y = _____

22. Complete the proof. Given: p ∥ q. Prove: ∠1 ≅ ∠4. (5)

Do NOT use the Alternate Interior Angles Theorem.



Statements	Reasons
1. p ∥ q	**1.** Given
2.	**2.**
3.	**3.**
4.	**4.**